

Appendix M: Glossary

Appendix M: Comprehensive Glossary of CNC Engineering Terms

M.1 Mechanical Design Terms

- **Backlash:** Lost motion in mechanical systems due to clearances between components (measured in mm or inches). Target: <0.05mm for precision CNC. Anti-backlash compensation: ball screw preload, spring-loaded nuts.
- **Ball Screw:** Precision lead screw using recirculating ball bearings to convert rotary to linear motion with high efficiency (90-95%) and low backlash (<0.01mm). Grades: C10 (+/-52mm/300mm) to C2 (+/-8mm/300mm).
- **Bearing Types:**
 - **Angular Contact:** High-speed spindle bearings with contact angle 15-40°. Preload required.
 - **Deep Groove Ball:** General-purpose radial/thrust bearing. Most common type.
 - **Tapered Roller:** Heavy radial and thrust loads. Used in large gantries.
 - **Linear Ball Bearing:** Recirculating ball bearing for linear motion on hardened shafts.
- **Cantilever:** Beam supported at one end only. Z-axis spindle head is cantilever load. Deflection = $FL^3/(3EI)$.
- **Compliance:** Inverse of stiffness (mm/N). Higher compliance = more deflection under load.
- **Critical Damping:** Damping ratio $\zeta=1$. System returns to equilibrium without oscillation. Under-damped: $\zeta<1$ (oscillates), Over-damped: $\zeta>1$ (slow return).
- **Datum:** Reference surface, line, or point from which measurements are made. Critical for alignment and repeatability. Kinematic datum: 3-2-1 constraint principle (6 DOF constraint).
- **Degrees of Freedom (DOF):** Number of independent motions. Rigid body: 6 DOF (3 translation + 3 rotation). CNC mill: 3 DOF (XYZ), 5-axis: 5 DOF.
- **Dynamic Load Rating (C):** Load capacity of bearing/guide that allows 1 million meters of travel (or 1 million revolutions for rotary bearings) before fatigue failure (90% reliability, L10 life).
- **Elastic Modulus (Young's Modulus):** Material stiffness (stress/strain ratio). Steel: 200 GPa, Aluminum: 69 GPa, Cast Iron: 100-140 GPa.
- **Fastener Preload:** Tensile force applied during bolt tightening. Prevents joint separation and improves fatigue life. Target: 75-90% of proof load.
- **Gantry:** Bridge structure spanning machine bed, carrying Y-axis and Z-axis. Types: open (one rail) vs. closed (two rails). Closed gantry: better torsional rigidity.
- **Hysteresis:** Lag between input and output in mechanical systems. Ball screw hysteresis: <2mm typical. Caused by micro-slip in contacts.

- **Inertia (Moment of):** Resistance to rotational acceleration. $I = mr^2$ for point mass. Critical for motor sizing and resonance analysis.
- **Kerf:** Width of material removed by cutting process. Plasma: 1-3mm, Fiber Laser: 0.1-0.5mm, Waterjet: 0.5-1.5mm, Router: tool diameter + deflection.
- **Kinematic Coupling:** Exact-constraint mounting using 3 spheres on 3 V-grooves (or equivalents). Provides repeatable positioning <1mm. Used in precision fixtures.
- **Lead:** Linear travel per revolution of screw. Ball screw: 5mm, 10mm, 20mm typical. Lead != pitch for multi-start screws.
- **Linear Guide (Profiled Rail):** Precision bearing system with hardened rail and ball-bearing carriage. Accuracy: +/-5-15mm over 300mm. Load ratings: 5-200 kN.
- **Load Capacity:**
 - **Dynamic (C):** For moving loads, L10 life calculation.
 - **Static (C₀):** Maximum load at standstill without permanent deformation.
 - **Moment Loads (M):** Torque about carriage axes. Causes premature wear.
- **Natural Frequency:** Frequency at which structure vibrates freely. $f_n = (1/2\pi)\sqrt{k/m}$. Avoid excitation near f_n (resonance amplification).
- **Pitch:** Distance between screw threads. Ball screw: pitch x starts = lead. Rack & pinion: pitch circle diameter/# teeth.
- **Preload:** Internal force applied to bearing or screw to eliminate clearance and increase stiffness. Ball screw: 3-7% of dynamic load. Linear guide: light/medium/heavy preload options.
- **Rack and Pinion:** Linear drive converting pinion rotation to rack translation. Module 1.5-3mm typical. Backlash: 0.1-0.3mm (larger than ball screw).
- **Repeatability:** Ability to return to same position. Spec: +/-10mm typical for CNC. Better than accuracy (absolute position error).
- **Resolution:** Minimum increment of motion. Encoder: 0.1-1mm typical. Stepper: microstep size (e.g., 1/8 step = 0.025mm for 5mm lead).
- **Runout (TIR):** Total Indicator Reading - maximum deviation of rotating component from true circular path. Spindle target: <5 mm TIR at nose. Collet runout: <10 mm.
- **Static Load Rating (C₀):** Maximum load bearing/guide can support without permanent deformation (0.0001x bearing diameter deformation criterion).
- **Stiffness:** Resistance to deflection under load, measured in N/mm. Higher = better precision under cutting forces. System stiffness: series springs ($1/k_{total} = 1/k_1 + 1/k_2 + \dots$).
- **Straightness:** Maximum deviation from true straight line over specified length. Linear rail: 20mm/m typical. Measured with autocollimator or laser interferometer.
- **Structural Loop:** Load path from cutting tool to workpiece through machine structure. Shorter/stiffer loop = higher rigidity and accuracy.
- **Thermal Growth:** Dimensional change due to temperature. Aluminum: 23.6 mm/(m°C). Critical for precision machines - requires thermal compensation or cooling.
- **Tramming:** Squaring spindle axis perpendicular to table. Indicator sweep method. Target: <0.01mm over 300mm diameter.
- **Torsional Rigidity:** Resistance to twisting under torque. Hollow tube 4-8x better than solid shaft of same weight. Critical for gantry beams.

- **Whipping:** High-frequency vibration of unsupported screw section. Occurs above critical speed: $N_{crit} = (4.76 \times 10^6 \times d) / L^2$ (RPM, mm). Requires center support.
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M.2 Electrical and Control Terms

- **AC Servo Motor:** Brushless motor with encoder feedback for closed-loop control. Higher power density than steppers. Typical: 0.1-15 kW per axis.
- **Absolute Encoder:** Position sensor providing unique code for each position. Retains position after power loss. Resolution: 17-25 bit (130k-33M positions/rev).
- **Bus Voltage:** DC link voltage in servo drive (rectified AC). Typically 300-400V DC from 230V AC, 600-800V DC from 480V AC.
- **Commutation:** Switching current between motor windings to produce rotation. Stepper: external drive. Servo: internal with encoder feedback (field-oriented control).
- **Current Loop:** Innermost control loop in servo drive. Bandwidth: 1-3 kHz. Controls motor torque proportional to commanded current.
- **Duty Cycle:** Percentage of time component operates vs. rests. Servo peak torque: 30s duty cycle typical (3x continuous). Intermittent: cycles <10 min.
- **Electromagnetic Compatibility (EMC):** Ability of equipment to operate without EM interference. CE marking requires EMC testing per IEC 61000 series.
- **Electromagnetic Interference (EMI):** Unwanted electromagnetic radiation from electrical components interfering with signals. Mitigation: shielded cables, star grounding, ferrites, filters.
- **Encoder:**
 - **Incremental:** Outputs A/B pulses (position change). Requires homing. Z-index pulse per revolution. Typical: 2500-10,000 PPR.
 - **Absolute:** Direct position reading. Battery-backed or multi-turn (magnetic gearing). 17-bit = 131,072 counts/rev = 0.0027°.
 - **Resolver:** Analog rotary transformer. Robust but low resolution. Used in harsh environments.
- **EtherCAT:** Real-time Ethernet protocol for motion control. 1000 axes at 1 kHz cycle time. Beckhoff, Delta, many drives support.
- **Field-Oriented Control (FOC):** Advanced servo algorithm controlling flux and torque independently. Enables maximum torque/amp efficiency.
- **Grounding:** Connecting electrical circuits to earth potential. Types:
 - **Safety Ground:** Green/yellow wire, protects against shock.
 - **Signal Ground:** Common reference for signals. Star topology prevents ground loops.
 - **Chassis Ground:** Frame/enclosure ground. Bonds to safety ground.
- **Hall Effect Sensor:** Magnetic position sensor. Used for commutation in brushless motors (3 sensors, 60° apart). Binary output.
- **Holding Torque:** Maximum torque stepper motor can produce at standstill (0 RPM). Decreases with speed (torque curves).
- **Incremental Encoder:** See Encoder (Incremental). Requires limit switch homing on power-up. More common than absolute (lower cost).
- **Inductance:** Opposition to current change in coil. Motor inductance: 1-50 mH. Limits acceleration capability (voltage = $L \times di/dt$).

- **Inrush Current:** Initial surge current when powering equipment. Servo drive: 20-100x steady-state. Requires soft-start or pre-charge resistor.
- **Isolation (Optical):** Galvanic separation using LED/phototransistor. Isolates sensitive circuits from high-voltage/noisy circuits. Typical: 2500V isolation.
- **Jerk:** Rate of acceleration change (m/s^3). High jerk causes vibration. S-curve acceleration profiles limit jerk for smooth motion.
- **Latency:** Delay between command and response. Motion control: $<1\text{ms}$ required. Ethernet latency: 0.1-1ms depending on protocol.
- **Microstepping:** Dividing motor full steps into smaller increments ($1/2, 1/4, 1/8, 1/16, 1/32, 1/64, 1/128$). Smoother motion but lower torque and positional accuracy at high microstep ratios.
- **MOSFET:** Metal-Oxide-Semiconductor Field-Effect Transistor. Fast switching device in motor drives. Low on-resistance (R_{ds_on} : 5-50 mOhms).
- **Optical Isolation:** See Isolation (Optical). Protects control circuits from motor noise and voltage spikes.
- **PID Control:** Proportional-Integral-Derivative feedback controller. Standard servo tuning: K_p (position gain), K_i (integral - removes steady-state error), K_d (derivative - damping).
- **Position Loop:** Outermost servo control loop. Bandwidth: 10-100 Hz. Compares commanded vs. actual position (encoder), outputs velocity command.
- **Power Factor:** Ratio of real power to apparent power. Inductive loads (motors): 0.6-0.9 typical. Unity (1.0) is ideal. VFDs improve power factor.
- **PWM (Pulse Width Modulation):** Motor drive technique varying duty cycle to control average voltage/current. Frequency: 4-20 kHz typical (above audible range).
- **Regenerative Braking:** Converting motor kinetic energy back to electrical energy during deceleration. Requires braking resistor (dissipate) or bus sharing (return to supply).
- **Resolver:** See Encoder (Resolver). Analog rotary position sensor. Rugged but requires resolver-to-digital converter (RDC) chip.
- **RMS (Root Mean Square):** Effective value of AC waveform. Motor continuous ratings in RMS amps/volts. Peak = $\sqrt{2}$ x RMS for sine wave.
- **Servo:** Closed-loop motor system with encoder feedback for precise position/velocity control. Corrects for disturbances (cutting forces, friction).
- **Shielding:** Conductive layer around cable preventing EM radiation. Braid (80-95% coverage) or foil (100% coverage). Ground shield at ONE end only (avoid ground loops).
- **Slip:** Difference between synchronous speed and actual speed in AC induction motors. 2-5% at full load. Servo motors: zero slip (synchronous).
- **Stepper Motor:** Open-loop motor moving in discrete steps. Holding torque: 0.3-30 N*m. Steps/rev: 200 (1.8°) or 400 (0.9°) typical. Loses steps if overloaded.
- **Transistor Ratings:**
 - **V_{ds}:** Drain-source voltage rating (e.g., 600V MOSFET).
 - **I_{ds}:** Continuous drain current (e.g., 50A MOSFET).
 - **R_{ds_on}:** On-resistance (e.g., 10 mOhms = 0.5W at 10A).
- **Trajectory Planning:** Calculation of position/velocity/acceleration profiles. Trapezoidal profile: constant accel/decel. S-curve: limited jerk.

- **Tuning:** Adjusting PID gains for optimal servo response. Manual (Ziegler-Nichols, Lambda), Auto-tune (drive feature). Goal: fast settling without overshoot.
 - **Velocity Loop:** Middle servo control loop. Bandwidth: 100-500 Hz. Compares commanded vs. actual velocity (encoder derivative), outputs current command.
 - **VFD (Variable Frequency Drive):** Electronic controller varying AC motor speed by adjusting frequency and voltage (V/Hz control). Spindle control: 0-400 Hz = 0-24,000 RPM.
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M.3 CNC Control and Programming Terms

- **Absolute Positioning (G90):** Coordinates specified relative to fixed origin (work zero). Example: G90 G01 X100 -> move to X=100.
- **Arc (Circular Interpolation):** G02 (CW) or G03 (CCW) circular path. Parameters: I/J (center offset) or R (radius). Requires XY plane (G17).
- **Automatic Tool Changer (ATC):** Mechanism swapping tools automatically. Carousel type: 8-40 tools. Arm type: 10-20s tool change. ISO 30/40/50 taper.
- **Axis Naming:**
 - **Primary (X, Y, Z):** Linear translation axes. Z typically vertical (spindle axis). Right-hand rule.
 - **Rotary (A, B, C):** Rotation about X, Y, Z respectively. 4th axis: rotary table. 5-axis: A+C or B+C.
- **Canned Cycles:** Pre-programmed routines for common operations. G81 (drilling), G83 (peck drilling), G84 (tapping), G73 (high-speed peck).
- **Compensation:**
 - **Tool Length (G43):** Offset Z position by tool length in table. G43 H01 = use tool 1 length.
 - **Tool Radius (G41/G42):** Offset path left/right by tool radius. Climb milling: G41. Conventional: G42.
 - **Backlash Compensation:** Software correction for mechanical backlash. LinuxCNC: [AXIS_0]BACKLASH parameter.
- **Contour Error (Following Error):** Difference between commanded and actual position during motion. Servo: <0.1mm typical. Exceeding limit triggers alarm.
- **Coordinate System:**
 - **Machine (G53):** Absolute position relative to home switches. Fixed reference.
 - **Work (G54-G59):** User-defined origins for parts. G54 most common. G10 L2 sets offsets.
- **Dry Run:** Test mode running program at reduced speed without tool engaged. Verify program before cutting.
- **DXF (Drawing Exchange Format):** CAD file format. 2D vector graphics. Common CAM input format for plasma/laser/waterjet.
- **Feed Rate:** Speed of tool movement through material (mm/min or in/min). Programmed with F-word. Higher = faster machining but more tool wear. Inverse time (G93) for 5-axis.
- **Feed Per Tooth (Chipload):** Material removed per cutter tooth. Formula: $f_z = F / (N \times \text{RPM})$ where F=feedrate, N=# teeth. Aluminum: 0.05-0.15 mm/tooth.
- **G-Code:** Machine programming language (RS-274). G00 (rapid), G01 (linear), G02/G03 (arc), G90/G91 (absolute/incremental).

- **HAL (Hardware Abstraction Layer):** LinuxCNC real-time software layer connecting G-code to hardware. Configuration: .hal files. Signals: bit/float/s32/u32 types.
- **Home Switch:** Limit switch establishing machine origin. Homing sequence: rapid to switch, slow off switch, zero position. Index pulse optional (repeatable to 0.1mm).
- **Incremental Positioning (G91):** Coordinates specified relative to current position. Example: G91 G01 X10 -> move 10mm from current X.
- **Interpolation:** CNC calculating intermediate points between programmed endpoints. Linear (G01): straight line. Circular (G02/G03): arc. NURBS: complex curves.
- **Inverse Kinematics:** Calculating joint positions for desired tool position. Required for robotic arms, delta robots. LinuxCNC: gensekins, pumakins.
- **Jog Mode:** Manual axis movement via buttons/handwheel. Continuous (hold button) or incremental (0.001", 0.01", 0.1" steps).
- **Look-Ahead:** Controller analyzing upcoming G-code to optimize velocity. Prevents overshoot at corners. Depth: 100-1000 segments typical.
- **M-Code:** Auxiliary machine functions (not motion). M03 (spindle CW), M08 (coolant on), M06 (tool change), M30 (program end).
- **Machine Coordinates:** Absolute position relative to machine home (physical limits). G53 ignores work offsets. Used for tool changes, safe positions.
- **MDI (Manual Data Input):** Keyboard entry of single G-code commands. Useful for setup, probing, jogging to position.
- **Modal:** Command remains active until cancelled. G01 (linear mode), G90 (absolute mode). Non-modal: G04 (dwell), G28 (home).
- **Offsets:**
 - **Work Coordinate (G54-G59):** Part origin locations. G10 L2 P1 X10 Y20 = set G54 to X+10, Y+20.
 - **Tool Length (G43):** Z compensation for tool length. Measured from gage line to tool tip.
 - **Tool Diameter (G41/G42):** Radius compensation around programmed path.
- **Override:** Real-time adjustment of programmed values. Feed override: 0-200% typical. Spindle override: 50-120%. Lets operator fine-tune.
- **Parametric Programming:** Using variables in G-code. #1=10 (assignment), G01 X#1 (use variable). Loops with WHILE, conditionals with IF.
- **Post-Processor:** Software converting CAM toolpaths to machine-specific G-code. Custom posts for each control (Fanuc, Haas, LinuxCNC, Mach3).
- **Probe (Touch Probe):** Precision switch for automatic workpiece/tool measurement. Renishaw: repeatability 0.5mm. G38.2 (probe toward), G38.4 (probe away).
- **Program Stop:** M00 (mandatory stop), M01 (optional stop - requires operator enable). Spindle/coolant off, requires cycle start to continue.
- **Rapid (G00):** Non-cutting move at maximum feedrate. Not guaranteed straight line (may move axes sequentially). No F-word.
- **Real-Time:** System guaranteeing response within specified time. LinuxCNC: 1ms servo cycle, 25µs base thread (steppers). RTAI or Preempt-RT kernel.
- **Scaling (G51):** Proportionally increase/decrease programmed dimensions. G51 X2 Y2 = double size. G50 cancels. Useful for families of parts.

- **Single Block:** Execution mode pausing after each G-code line. Used for program verification and debugging.
 - **Spindle Synchronization:** Coordinating spindle rotation with linear motion. Rigid tapping (G33.1): spindle encoder required. Thread milling: G02/G03 with helical interpolation.
 - **Subprogram:** Reusable G-code block called with M98 P1234 (call program O1234). M99 returns. Useful for repeated patterns (bolt holes).
 - **Tool Offset:** Compensation for tool length/diameter differences. Length offset (G43): Z-axis. Radius compensation (G41/G42): XY plane. Stored in tool table.
 - **Tool Table:** Database of tool dimensions. Tool # (T), length offset (H), diameter (D), pocket (P). G10 L1 sets values. G43 H01 applies tool 1 length.
 - **Trajectory Planner:** Calculates smooth motion profiles respecting velocity/acceleration limits. Acceleration: 0.5-5 m/s² typical. Jerk limiting optional.
 - **Work Offset (G54-G59):** User-defined origin for part coordinates. G54 most common (work offset 1), allows multiple fixtures/setups. Extended: G54.1 P1-P9 (G59.3).
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M.4 Machining and Cutting Terms

- **Abrasive Waterjet (AWJ):** Waterjet with garnet abrasive (80 mesh typical). Cuts metals, stone, ceramics. Pressure: 55,000-90,000 PSI. Kerf: 0.8-1.5mm.
- **Adaptive Clearing:** CAM toolpath maintaining constant tool engagement. Varies feedrate and stepover for consistent chip load. Reduces tool wear 30-50%.
- **Built-Up Edge (BUE):** Material welding to tool edge at low speeds. Causes poor finish and tool failure. Solution: increase speed or use coated tools.
- **Chip Evacuation:** Removing chips from cutting zone. Methods: flood coolant (washing), air blast, vacuum, chip conveyor. Poor evacuation causes recutting and heat.
- **Chipload:** Thickness of chip removed per tooth. Formula: Chipload = Feed / (RPM x # teeth). Aluminum roughing: 0.10-0.20 mm/tooth. Finishing: 0.03-0.08 mm/tooth.
- **Chatter:** Self-excited vibration causing poor surface finish and tool breakage. Frequency: 100-5000 Hz. Causes: low rigidity, excessive DOC/WOC, tool runout. Solutions: reduce DOC, change RPM, increase rigidity.
- **Climb Milling (Down Milling):** Cutter rotation direction matches feed direction. Chip thickness: thick to thin. Advantages: better finish, lower forces, longer tool life. Requires backlash elimination.
- **Conventional Milling (Up Milling):** Cutter rotation opposes feed direction. Chip thickness: thin to thick. Advantages: works with backlash. Disadvantages: lifts workpiece, more tool wear.
- **Coolant:** Fluid for cooling tool and workpiece. Types: water-soluble (emulsion, 5-20:1 dilution), synthetic, straight oil, MQL. Functions: cooling, lubrication, chip evacuation.
- **Cutter Compensation:** Tool radius offset (G41/G42). Programming to part edge, controller offsets by tool radius. Enables tool wear compensation without reprogramming.
- **Cutting Force:** Resistance of material to cutting. Components: tangential (Ft), radial (Fr), axial (Fa). $F_t = K_s \times \text{DOC} \times \text{WOC} \times f_z^n$ (specific cutting force Ks from tables).
- **Cutting Speed (Vc):** Velocity at tool cutting edge. Formula: $V_c = \pi \times D \times N$ where D=diameter (mm), N=RPM. Steel: 100-300 m/min HSS, 150-500 m/min carbide.
- **Depth of Cut (DOC):** Axial depth tool enters material. Symbol: ap. Roughing: 3-10mm or 50-100% of tool diameter. Finishing: 0.1-0.5mm.

- **EDM (Electrical Discharge Machining):** Material removal by electrical sparks. Wire EDM: 0.1-0.3mm wire cuts intricate shapes. Sinker EDM: shaped electrode. Surface finish: 0.1-3µm Ra.
- **Feeds and Speeds:** Feed rate (mm/min or IPM) and spindle speed (RPM) for specific material/tool combination. Calculate from chipload, # teeth, cutting speed.
- **Fixture:** Device holding workpiece during machining. Types: vise, toe clamps, vacuum table, custom (3-2-1 locating). Rigidity critical - workpiece stiffness > machine stiffness.
- **Flood Coolant:** High-volume coolant delivery (5-50 GPM). Nozzles at tool-workpiece interface. Removes heat and chips. Requires sump, pump, filtration.
- **Garnet (Abrasive):** Crushed mineral for abrasive waterjet cutting. Mesh size: 50 (coarse), 80 (standard), 120 (fine). Cost: (0.10-)0.30/lb. Consumption: 0.3-1.0 lb/min.
- **Heat-Affected Zone (HAZ):** Region near cut edge with altered microstructure. Plasma/laser: 0.5-2mm HAZ. Waterjet/router: zero HAZ (cold cutting).
- **High-Speed Machining (HSM):** Cutting at 5-10x conventional speeds. Aluminum: 15,000-30,000 RPM. Steel: 8,000-15,000 RPM. Requires rigid machine, balanced tools, high-performance spindle.
- **Kerf:** Width of material removed by cutting process. Compensation: program 0.5x kerf inside (internal features) or outside (external features).
- **Lead-In/Lead-Out:** Ramp or arc entry/exit to cut to avoid plunge marks. Circular lead-in: radius 1-5mm. Helps tool engagement and surface finish.
- **Material Removal Rate (MRR):** Volume of material removed per unit time. $MRR = DOC \times WOC \times \text{Feed}$ (mm³/min). Roughing: maximize MRR. Finishing: minimize (better surface).
- **Minimum Quantity Lubrication (MQL):** Micro-mist lubrication (0.01-0.1 mL/hr oil + air). Replaces flood coolant. Benefits: dry chips (recyclable), no disposal, cleaner. Aluminum, cast iron applications.
- **Nesting:** Arranging parts on material sheet to minimize waste. CAM nesting software. Common spacing: 3-10mm between parts (avoid thermal effects in plasma/laser).
- **Pierce:** Initial penetration of material before cutting. Plasma: 2-5 seconds pierce time, height above work (pierce height = 2x cut height). Laser: oxygen assist for thick steel.
- **Power (Spindle):** $P = (K_s \times DOC \times WOC \times \text{Feed}) / (60 \times \eta)$ where K_s =specific cutting force (N/mm²), η =efficiency (0.7-0.9). Steel: 0.5-10 kW typical.
- **Radial Immersion:** Percentage of cutter diameter engaged in material. Full slot: 100%. Half slot: 50%. Light finish: 10-30%. Affects cutting forces and tool life.
- **Roughing:** High-MRR material removal leaving finishing stock. Larger DOC/WOC, faster feed, 0.1-0.5mm stock remaining. Surface finish: 3-10 µm Ra.
- **Scallop Height:** Residual cusps between toolpath passes. Smaller stepover = smaller scallop. Formula: $h = \text{stepover}^2 / (8R)$ for ball end mill radius R.
- **Speeds and Feeds:** See Feeds and Speeds. Critical for tool life and surface finish. Too fast: excessive heat/wear. Too slow: built-up edge, poor finish.
- **SFM (Surface Feet per Minute):** Cutting speed at tool edge (imperial). Formula: $SFM = (\pi \times D \times RPM) / 12$ (D in inches). Metric equivalent: V_c (m/min).
- **Stepover:** Lateral offset between toolpath passes. Percentage of tool diameter. Roughing: 40-60%. Finishing: 5-20%. Smaller = better finish, longer cycle time.
- **Surface Finish (Ra):** Arithmetic average roughness. Typical: Roughing 3-10 µm, General 0.8-3 µm, Precision 0.2-0.8 µm, Grinding <0.2 µm. Measured with profilometer.

- **Through-Spindle Coolant (TSC):** High-pressure coolant (50-100 PSI) through hollow spindle and tool. Essential for deep hole drilling (>3x diameter). Rotary union seals coolant to rotating spindle.
 - **Trochoidal Milling:** Circular toolpath with advancing motion. Maintains small radial engagement (10-20%). Benefits: longer tool life, higher MRR in hard materials, deeper cuts.
 - **Width of Cut (WOC):** Radial depth (stepover in one pass). Symbol: ae. Slotting: 100% diameter. Roughing: 40-60% diameter. Finishing: 10-30% diameter.
 - **Work Hardening:** Material surface hardening from plastic deformation. Austenitic stainless (304/316): severe work hardening. Requires positive rake tools and adequate feedrates (avoid rubbing).
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M.5 Material and Metallurgy Terms

- **Alloy:** Metal mixture improving properties. Steel: C + Fe + Mn/Cr/Ni. Aluminum: 6061 (Mg+Si), 7075 (Zn). Properties vary widely by composition.
- **Annealing:** Heat treatment reducing hardness and internal stress. Process: heat to 700-900°C, slow cool. Makes material easier to machine.
- **Austenite:** FCC crystal structure of iron (stable above 912°C). Non-magnetic. Retained in stainless 304/316 (austenitic stainless).
- **Carbide (Cemented Carbide):** WC particles in Co binder. Hardness: 1500-2000 HV (vs. 800 HV HSS). Cutting tools: grades C1-C8 (ISO K, M, P). Brittle but heat-resistant.
- **Case Hardening:** Surface hardening via carburizing/nitriding. Hard exterior (50-62 HRC), tough core (30-40 HRC). Gears, bearing races.
- **Cast Iron:** Fe-C alloy with 2-4% carbon. Types: Gray (graphite flakes, good damping), Ductile (nodular graphite), White (cementite, very hard). Machinable without coolant.
- **Ferrite:** BCC crystal structure of iron (stable <912°C). Soft, magnetic. Pure iron mostly ferrite at room temperature.
- **Grain Size:** Crystal size in material. Finer grain = stronger (Hall-Petch effect). Coarse grain: easier to machine but weaker. ASTM grain size number: 5-8 typical.
- **Hardness:** Resistance to deformation. Scales: Rockwell (HRC 20-65), Brinell (HB 100-650), Vickers (HV 100-1000). Conversion tables available. Harder materials require lower feeds/speeds.
- **Heat Treatment:** Controlled heating/cooling altering microstructure. Types: annealing (soften), normalizing (refine grain), quenching (harden), tempering (toughness).
- **High-Speed Steel (HSS):** Tool steel with W, Mo, Cr, V. Hardness: 62-65 HRC. Cutting tools: drills, taps, end mills. Lower cost than carbide but lower heat resistance.
- **Martensite:** Hard, brittle phase formed by rapid quenching steel from austenite. BCT structure. 58-65 HRC typical. Requires tempering to reduce brittleness.
- **Precipitation Hardening:** Strengthening via precipitate formation. Aluminum 6061-T6: Mg₂Si precipitates. Stainless 17-4PH: copper precipitates. Age hardening process.
- **Quenching:** Rapid cooling from austenite temperature forming martensite. Media: water (fastest, cracking risk), oil (moderate), air (slow). Hardness depends on cooling rate.
- **Rockwell Hardness (HRC):** Hardness scale using diamond cone indenter. HRC 20 (mild steel), HRC 30-40 (prehardened steel), HRC 50-60 (hardened steel, cutting tools). Converted to HB, HV.
- **Stainless Steel:** Fe-Cr alloy (>10.5% Cr) forming passive oxide layer. Types:

- **Austenitic (304, 316):** Non-magnetic, corrosion-resistant, work-hardens severely. Difficult to machine.
 - **Ferritic (430):** Magnetic, less corrosion resistance, easier to machine.
 - **Martensitic (410, 420):** Hardenable, magnetic, moderate corrosion resistance.
 - **Temper:** Designation of aluminum heat treatment. T6: solution treated + artificially aged (peak strength). T4: solution treated + naturally aged. O: annealed (soft).
 - **Tempering:** Reheating quenched steel to reduce brittleness and stress. Temperature: 150-650°C. Higher temp = softer but tougher. Required after quenching to avoid cracking.
 - **Thermal Conductivity:** Heat transfer rate. Aluminum: 205 W/(mK), Steel: 50 W/(mK), Stainless: 15 W/(mK). Higher = better coolant effectiveness, less thermal damage.
 - **Toughness:** Energy absorption before fracture. Opposite of brittleness. Charpy/Izod impact tests. Tempered martensite: tough. Untempered martensite: brittle.
 - **Ultimate Tensile Strength (UTS):** Maximum stress before failure. Mild steel: 400 MPa. 4140 steel: 655 MPa (annealed) to 1800 MPa (quenched). Design safety factor: 2-4x.
 - **Yield Strength:** Stress causing permanent deformation (0.2% offset). Design limit for structures. Mild steel: 250 MPa. Aluminum 6061-T6: 276 MPa. Factor of safety applied.
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M.6 Safety and Regulatory Terms

- **Arc Flash:** Explosive electrical fault creating plasma arc. Energy: up to 40 cal/cm² (lethal). Protection: arc-rated PPE, proper clearances, lockout/tagout.
- **ANSI (American National Standards Institute):** Standards organization. B11 series: machine tool safety. Z87.1: eye protection. Z49.1: welding safety.
- **CE Marking:** EU declaration of conformity to safety directives. Required for machinery sold in Europe. Self-certification or notified body assessment.
- **Confined Space:** Area with limited egress and not designed for continuous occupancy. Sump, tank interiors. OSHA 1910.146: permit required, atmosphere testing, rescue plan.
- **Double Insulation:** Two independent layers of insulation protecting against shock. Class II equipment (no ground required). Symbol: square within square.
- **E-Stop (Emergency Stop):** Immediately removes power to motion/spindle. Required by ISO 13850 on all machines. Type: Category 0 (uncontrolled stop) or Category 1 (controlled stop then power off). Red mushroom button with yellow background.
- **Enabling Device:** Three-position switch for pendant operation. Squeezed: machine enabled. Fully pressed or released: machine disabled. Prevents unintended motion.
- **Fall Protection:** Arrest system for work >4 feet height (OSHA). Full-body harness + lanyard + anchor. Machine access platforms require guardrails or personal fall arrest.
- **Ground Fault Circuit Interrupter (GFCI):** Detects current imbalance (leakage to ground) and trips circuit. 5 mA threshold, <25ms trip time. Required for temporary/outdoor 120V circuits.
- **Guarding:** Physical barrier preventing access to hazards. Fixed (permanent), interlocked (door switch), adjustable. ANSI B11.19 specifies distances and openings.
- **Hazard Analysis:** Systematic identification of dangers. Risk = Severity x Probability. ISO 12100 risk assessment. Mitigation hierarchy: eliminate, guard, warn, train, PPE.
- **Interlock:** Safety device preventing machine operation unless conditions met (door closed, guards in place). Positive opening (mechanically driven to open state). Monitored by safety relay/PLC.

- **ISO (International Organization for Standardization):** Global standards body. ISO 12100: machinery safety principles. ISO 13849-1: safety control systems. ISO 13850: emergency stop.
- **Light Curtain:** Optical safety device creating invisible barrier. 14mm spacing between beams typical. Safety category 2 or 4. Response time: 10-30ms. Replaces physical guarding.
- **Lockout/Tagout (LOTO):** Procedure isolating energy sources during maintenance (OSHA 1910.147). Lock device on disconnects, tags identifying worker. De-energize, dissipate stored energy, verify zero energy.
- **Machine Guarding:** See Guarding. OSHA 1910.212 general requirements. Point of operation guarding: distance to hazard vs. opening size tables (1/4" opening = 4" distance minimum).
- **NFPA (National Fire Protection Association):** Fire safety standards. NFPA 70 (National Electrical Code), NFPA 79 (Industrial Machinery Electrical Standard), NFPA 654 (Combustible Dust).
- **OSHA (Occupational Safety and Health Administration):** US workplace safety enforcement. 29 CFR 1910: General Industry standards. 1910.212: machine guarding, 1910.147: LOTO, 1910.1000: air contaminants.
- **PEL (Permissible Exposure Limit):** Maximum airborne concentration of substance allowed (OSHA). Oil mist: 5 mg/m³ (8-hr TWA). Hexavalent chromium (welding stainless): 5 µg/m³.
- **Performance Level (PL):** ISO 13849-1 safety reliability rating. PL_a (lowest) to PL_e (highest). PL_d/PL_e: redundant channels, diagnostics, fault tolerance. Emergency stops: typically PL_d.
- **PPE (Personal Protective Equipment):** Safety gear required for operation. Hierarchy: last resort (engineering controls preferred). Minimum CNC: Z87.1 safety glasses, hearing protection >85 dB, steel-toe boots.
- **Residual Current Device (RCD):** European term for GFCI. 30 mA threshold typical for personnel protection. 300 mA for equipment protection (fire prevention).
- **Risk Assessment:** Evaluating hazard severity and probability. ISO 12100 methodology. Risk matrix: severity (1-4) x probability (1-4) = risk (1-16). Acceptable risk: <6 typical.
- **Safety Category:** IEC 60204-1 classification of emergency stop reliability. Category 0: uncontrolled stop (immediate power removal). Category 1: controlled stop, then power removal. Category 2: controlled stop, power maintained.
- **Safety Integrity Level (SIL):** IEC 61508 classification of safety system reliability. SIL 1 (10⁻¹ to 10⁻² failure probability) to SIL 4 (10⁻⁵ to 10⁻⁴). Industrial machinery: typically SIL 2.
- **Safety Relay:** Redundant relay module monitoring safety circuits. Dual-channel inputs, force-guided contacts (mechanically linked NO/NC). Pilz, Phoenix Contact manufacturers.
- **Two-Hand Control:** Operator must use both hands on buttons to activate machine. Prevents hands in hazard zone. Anti-tie-down: buttons <300mm apart, must press within 0.5s.

M.7 Measurement and Metrology Terms

- **Accuracy:** Closeness of measurement to true value. Specification: +/-10 µm/300mm typical for CNC. Systematic error (calibrated out with compensation).
- **Autocollimator:** Optical instrument measuring angular deviation. Resolution: 0.1 arc-second (0.5 µm/m). Leveling precision machines, measuring spindle axis alignment.
- **Calibration:** Comparison to known standard. Traceable to NIST/NPL national standards. Period: annual for production equipment, quarterly for metrology. Adjustment if out of tolerance.

- **Coordinate Measuring Machine (CMM):** Precision measurement device with touch probe on 3-axis gantry. Accuracy: 2-5 μm typical. Temperature-controlled room ($20^{\circ}\text{C} \pm 1^{\circ}\text{C}$).
- **Dial Indicator:** Contact probe with 0.001" (0.01mm) resolution. Long-range (1-3"), continuous dial. Used for setup, runout measurement, tramping spindle.
- **Feeler Gauge:** Precision thickness blades for measuring gaps. 0.001" (0.025mm) to 0.040" (1mm) typical set. Used for gap setting, workpiece leveling.
- **Gauge Block (Jo Block):** Precision length standard. Grade 0: $\pm 0.05 \mu\text{m}$, Grade 1: $\pm 0.1 \mu\text{m}$, Grade 2: $\pm 0.2 \mu\text{m}$. Wrung together to create exact lengths. Calibration reference.
- **Interferometry (Laser):** Optical measurement using light wave interference. Accuracy: $<1 \text{ ppm}$ (1 $\mu\text{m}/\text{m}$). Renishaw XL-80 laser: 0.5 ppm linear accuracy. Machine tool calibration.
- **Micrometer:** Precision screw-threaded measuring instrument. 0.001" (0.01mm) graduation, 0.0001" (0.001mm) vernier. Outside, inside, depth types. Thimble friction ratchet for consistent force.
- **Parallax Error:** Measurement error from viewing dial at angle. Avoided by perpendicular viewing or mirror scale (align pointer with reflection).
- **Precision:** Repeatability of measurements (scatter). Standard deviation of repeated measurements. High precision with poor accuracy: systematic error (bias).
- **Repeatability:** Ability to return to same position. Spec: $\pm 10 \mu\text{m}$ typical for CNC. Better than accuracy (absolute position error). Critical for production machining.
- **Resolution:** Minimum increment of motion or measurement. Encoder: 0.1-1 μm typical. Stepper: microstep size. Display resolution $\neq$ accuracy.
- **Straightedge:** Precision flat reference. Grade A: 50 $\mu\text{m}/\text{m}$, Grade AA: 15 $\mu\text{m}/\text{m}$. Camelback beam shape (self-weight sag compensation). Used with feeler gauges, height gauges.
- **Surface Plate:** Precision flat reference surface. Granite (Grade A, AA, AAA). Grade A: 25 μm over 600x900mm plate. Temperature-stable (20°C). Inspection and layout.
- **Test Indicator (Dial Test Indicator):** Horizontal plunger indicator with 0.0001"-0.001" graduation. Small contact point. Used for fine centering, edge finding, small radii.
- **Tolerance:** Acceptable variation from nominal dimension. $\pm 0.1\text{mm}$ typical. ISO 2768: general tolerances (fine, medium, coarse). Geometric tolerances: position, flatness, perpendicularity (GD&T).
- **Traceability:** Documented chain to national standards. Calibration certificate from NIST-accredited lab. Required for AS9100 aerospace, ISO 17025 metrology labs.
- **Vernier Scale:** Secondary scale for interpolation between main scale graduations. 0.02mm or 0.001" resolution typical. Used on calipers, height gauges, protractors.

M.8 Plasma, Laser, and Waterjet Terms

- **Abrasive Waterjet (AWJ):** See Waterjet terminology. Garnet cutting hard materials (steel, titanium, stone, glass).
- **Assist Gas:** Gas flow during laser cutting improving quality and speed. Oxygen (steel: exothermic oxidation), Nitrogen (stainless, aluminum: inert, no oxidation), Air (mild steel, budget option).
- **Beam Divergence:** Laser beam spreading with distance. $<0.1 \text{ mrad}$ typical for fiber lasers. Affects focal position sensitivity and cut edge taper.
- **Consumables (Plasma):** Electrode, nozzle, swirl ring, shield cap. Life: 1-4 hours piercing heavy plate, 10-50 hours cutting thin sheet. Cost: (5-)30 per set depending on amperage.

- **Cutting Height:** Nozzle standoff from workpiece. Plasma: 3-6mm (initial height controllers maintain). Laser: focal point at or below surface. Waterjet: 2-5mm (abrasive jet spread).
- **Dross:** Solidified slag on cut edge underside. Plasma/laser: oxidized metal. Minimized by proper parameters (speed, gas flow, height). Removal: grinding, wire wheel.
- **Focal Length:** Distance from laser lens to focal point. Common: 5" (127mm), 7.5" (190mm). Shorter = smaller spot, thinner material. Longer = more working distance, less sensitive.
- **Heat-Affected Zone (HAZ):** See Machining Terms. Plasma: 1-2mm HAZ with hardness changes. Laser: 0.3-1mm HAZ. Waterjet: zero HAZ (cold process).
- **Hypertherm:** Leading plasma system manufacturer. PowerMax (handheld), Powermax (light duty), HPR (heavy duty). Market standard for CNC plasma.
- **Kerf Width:** See General terms. Plasma: 1-3mm depending on amperage (45A-200A). Laser: 0.1-0.5mm (1-6 kW fiber). Waterjet: 0.8-1.5mm (orifice + abrasive expansion).
- **Nozzle (Plasma):** Copper component focusing plasma arc. Orifice size: 0.8-2mm depending on amperage. Cools rapidly during cutting. Replace when orifice eroded >25%.
- **Orifice (Waterjet):** Sapphire or diamond jewel forming high-pressure water stream. Diameter: 0.010"-0.020" (0.25-0.50mm). Life: sapphire 50-100 hrs, diamond 800-2000 hrs.
- **Pierce (Plasma/Laser):** Initial penetration before cutting. Plasma: 1-5 seconds at 1.5-2x cut height to avoid blowback. Laser: 0.1-2 seconds depending on power and thickness.
- **Power Density:** Laser power per unit area. 10^6 - 10^8 W/cm² for cutting. Higher = faster cutting speeds. Focused fiber laser: 10x CO₂ laser power density.
- **Preheat (Plasma):** Lower-amperage arc stabilization before main arc transfer. 10-30A preheat. Improves arc starts and consumable life.
- **Taper (Cut Edge):** Angle deviation from vertical. Plasma: 0-3° typical. Laser: 0-2°. Waterjet: trailing edge (jet lag). Minimized by slow speeds and proper standoff.
- **THC (Torch Height Control):** Automated nozzle height adjustment during cutting. Capacitive sensor (non-contact) or arc voltage (plasma). Maintains constant height over warped sheets.
- **Transfer Arc:** Main plasma cutting arc (vs. pilot arc). Workpiece completes electrical circuit. Higher efficiency than non-transfer arc (gouging).
- **Wavelength:** Laser light wavelength. CO₂: 10.6 μ m (infrared). Fiber/Nd:YAG: 1.06 μ m (near-infrared). Fiber absorbed better by metals (higher cutting efficiency).

End of Glossary